Module 41.10: Comparative Study: Sequence Processing with RNNs vs Spatial Convolutions with CNNs

1. Connecting NLP Sentiment Analysis with Vision Modeling

Throughout Lectures 39, 40, and 41, we explored two distinct deep learning architectural paradigms:

1. **Convolutional Neural Networks (CNNs):** Designed for grid-structured spatial data (e.g., CIFAR-10, ImageNet).
2. **Recurrent Neural Networks (RNNs):** Designed for time-dependent, sequential data (e.g., IMDB text reviews, audio, time-series).

A classic educational question arises: **Can a recurrent neural network process visual images, and how does its inductive bias compare to a convolutional neural network?**

To answer this, we examine the canonical benchmark: **Handwritten Digit Classification on MNIST (28×28 grayscale pixels)** modeled through both spatial 2D convolution and temporal 1D recurrence (as highlighted in Lecture 41 reference materials `9 Handwritten Digit Classification using CNN & RNN.png` and `10 MNIST Dataset in PyTorch.png`).

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2. Notebook-Friendly Architecture Diagram

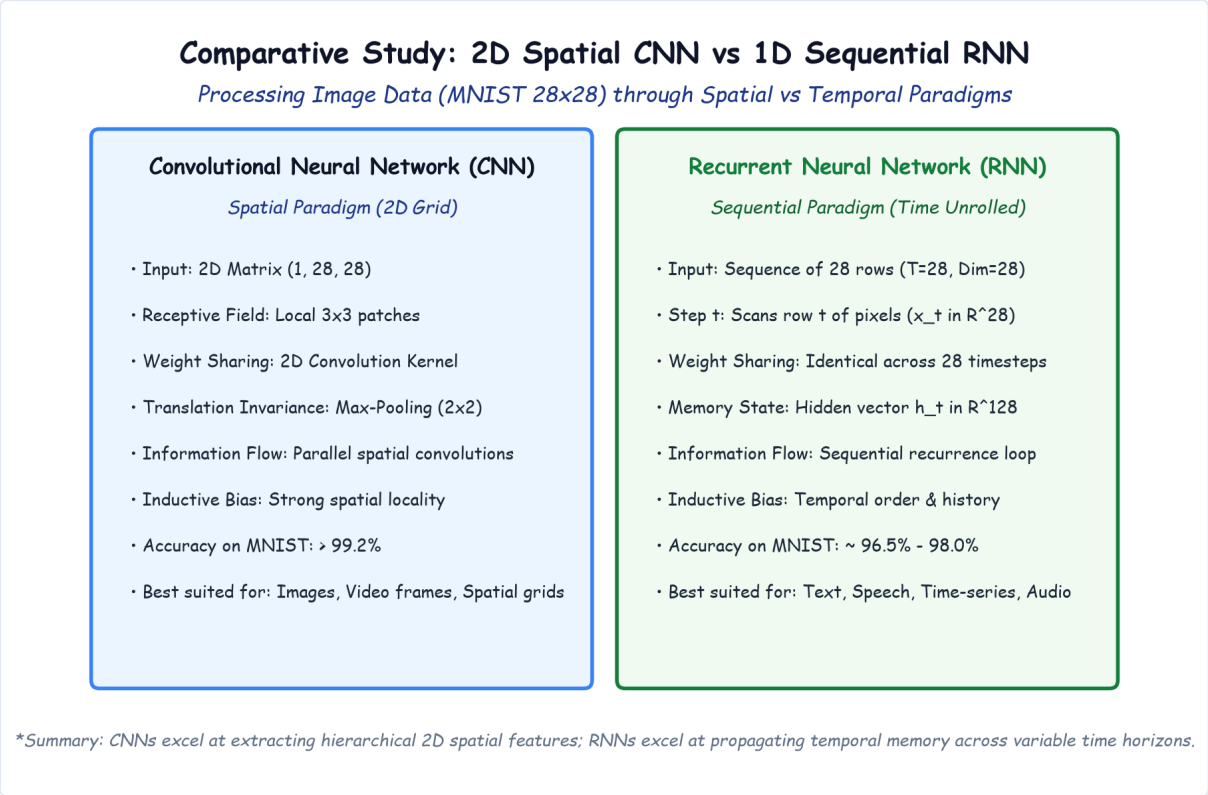


Figure 41.10: Architectural comparison of processing MNIST digit data: 2D Spatial Convolution (local receptive fields and translation invariance) vs 1D Sequential Recurrence (scanning 28 sequential rows of 28 pixels across time steps).

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3. The 2D Spatial Paradigm (CNN)

3.1 Mechanics

A Convolutional Neural Network views an image as a 2D spatial grid:

$$\mathbf{X} \in \mathbb{R}^{1 \times 28 \times 28}$$

- **Local Receptive Fields:** Kernels (e.g., 3×3 or 5×5) slide across both horizontal and vertical axes simultaneously.
- **Weight Sharing:** The same filter weights are applied across every spatial patch, encoding **translation equivariance**:

$$f(\text{shift}(\mathbf{x})) = \text{shift}(f(\mathbf{x}))$$

- **Spatial Pooling:** Max-pooling partitions feature maps, progressively halving spatial resolution ($28 \times 28 \to 14 \times 14 \to 7 \times 7$) while retaining dominant feature activations.

3.2 Performance & Inductive Bias

- **Inductive Bias:** Extremely strong prior that pixels close together in 2D space are correlated, regardless of absolute coordinates.
- **Accuracy on MNIST:** Typically $> 99.2\%$ with under $50,000$ parameters.

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4. The 1D Sequential Paradigm (RNN on Image Rows)

4.1 Mechanics

An image can be reinterpreted as a temporal sequence of row vectors unrolled across time:

- Total Sequence Length: $T = 28$ time steps.
- Input Dimension per Step: $\mathbf{x}_t \in \mathbb{R}^{28}$ (the 28 pixel values comprising row t).
- Full input tensor: $\mathbf{X} \in \mathbb{R}^{\text{Batch} \times 28 \times 28}$.

At time step $t = 1$, the RNN reads Row 1 (the top of the handwritten digit). At time step $t = 14$, it consumes Row 14, combining it with hidden state $\mathbf{h}_{13}$ carrying memory of previous rows. At step $t = 28$, the final hidden state $\mathbf{h}_{28}$ is projected to 10 class logits:

$$\mathbf{h}_t = \tanh(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h)$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_{hy}\mathbf{h}_{28} + \mathbf{b}_y)$$

4.2 Performance & Inductive Bias

- **Inductive Bias:** Strong assumption of 1D directional sequential dependency (top-to-bottom).
- **Limitations:** Completely blind to 2D local context (e.g., vertical spatial features spanning across non-adjacent rows must be propagated entirely through hidden memory).
- **Accuracy on MNIST:** Achieves $\approx 96.5\% - 98.0\%$. While surprisingly high, it is strictly inferior to CNNs due to the unnatural temporal formulation of 2D data.

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5. Architectural Comparison Matrix

Property	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Primary Domain	Computer Vision (Images, 2D Grids, Video)	Natural Language, Speech, Time-Series
Input Structure	2D/3D Tensor: (C, H, W)	1D Sequence Tensor: (T, d_x)
Computational Parallelism	**High** (all convolutions computed in parallel on GPU)	**Low** (sequential recurrence bottleneck: step t must wait for $t-1$)
Memory Horizon	Receptive field depth (stacking L layers)	Variable temporal memory (limited by vanishing gradients in vanilla RNN)
Key Advantage	Spatial translation invariance & parameter efficiency	Flexible variable-length sequential processing
Modern Evolution	ConvNeXt, Vision Transformers (ViT)	LSTMs, GRUs, and Attention-based Transformers